

Mathematical Intuition:- Dataset

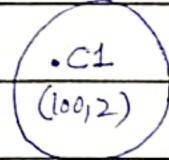
Point	Calories	Protein
A	100	2
B	120	3
C	110	1
D	300	25

* $k = 2$ (clusters)

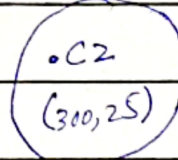
* Initialize centroids:-

Randomly initialize centroids.

cluster 1



cluster 2



* Calculate Euclidean Distance:-

from each point to each centroid.

$$= d(x, y) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

For C1,

$$A(100, 2) = \sqrt{(100 - 100)^2 + (2 - 2)^2} = 0.00$$

$$B(120, 3) = \sqrt{(100 - 120)^2 + (3 - 2)^2} = \sqrt{400 + 1} = \sqrt{401} = 20.02$$

$$C(110, 1) = \sqrt{(110 - 100)^2 + (1 - 2)^2} = \sqrt{100 + 1} = \sqrt{101} = 10.05$$

$$D(300, 25) = \sqrt{(300 - 100)^2 + (25 - 2)^2} = \sqrt{40000 + 529} = \sqrt{40529}$$

$$= 201.3$$

For C_2 ,

$$A(100, 2) = \sqrt{(100-300)^2 + (25-2)^2} = \sqrt{40000 - 529} = 201.3$$

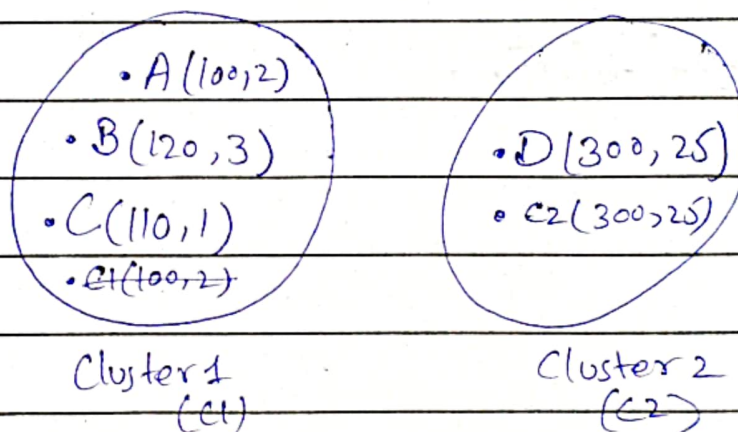
$$B(120, 3) = \sqrt{(120-300)^2 + (25-3)^2} = \sqrt{32400 + 484} = \sqrt{32884} = 181.3$$

$$C(110, 1) = \sqrt{(110-300)^2 + (1-25)^2} = \sqrt{36100 + 576} = \sqrt{36676} = 191.5$$

$$D(300, 25) = \sqrt{(300-300)^2 + (25-25)^2} = 0.00$$

* Assign to clusters:-

Assign the smallest distance point to the specific cluster.



* Update Centroids:-

Centroids of clusters update by taking mean of all the points.

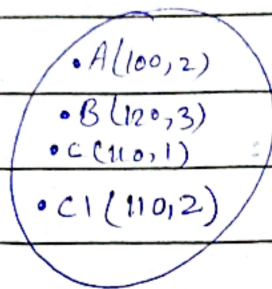
Cluster 1:

$$\text{new} = \left(\frac{100+120+110}{3}, \frac{2+3+1}{3} \right)$$

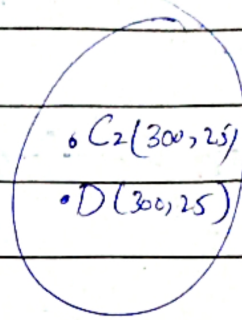
$$\text{new} = \left(\frac{330}{3}, \frac{6}{3} \right) = (110.0, 2.0)$$

Cluster 2:-

There is no need for updation because there is only one data point exist.



cluster 1



cluster 2

* Recalculate:-

Repeat the calculations with new centroids until convergence.

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When assignments don't change →
centroids won't move → Algorithm
has converged.